Nomenclature

n	mean orbital rate
a_i	control accelerations
e_i	positional error
$\dot{e}_i$	rate error
K_p	position gain
K_d	rate gain
ζ	damping ratio
ω_n	natural frequency
R	radius
μ	gravitational parameter
KOZ	keep-out-zone
TV	target vehicle
CV	chase vehicle
E	Earth

GFCS - GOLD Rules

1.30 The Attitude Control System (ACS) shall have stability margins of at least 6db for rigid body stability with 30 degrees phase margin. The magnitude of the flexible modes in the open-loop transfer function shall be less than minus 12dB.

- Proper gain and phase margins are required to maintain stability for reasonable unforeseen changes and uncertainty in spacecraft configuration.

1.31 The Attitude Control System (ACS) actuator sizing shall reflect specified allowances for mass properties growth.

- Knowledge of spacecraft mass and inertia can be very uncertain at early stages, so actuator sizing should be done with the appropriate amount of margin to ensure a viable design.

RPO Part 1 (Traveling to the KOZ)

Prompt

Using the Clohessy-Wiltshire (CWH) or Hill's equations to describe the relative motion of a chase vehicle (CV) approaching a target vehicle (TV) in orbit. MATLAB for computation and visualization. This is mean to simplify the problem instead of calculating the complexity of absolute orbits of both vehicles. CV is in a non-inertial coordinate frame fixed in the TV. The TV will Swift Observatory and the CV will represent SPEAR.

Reference Frame

- r (Radial-track): Parallel to the TV's radius vector from Earth.
- s (Along-track): Normal to the radial vector in the orbit plane.
- z (Cross-track): Perpendicular to the TV's orbital plane.

Assumptions

- TV's orbit is circular
  - Mean orbital rate is constant ($n \approx d\theta/dt$)
- CV and TV separation distance is much smaller than distance to Earth center

Discussion & Analysis

1. Development of a controller will be necessary to solve this problem; however, we are going to start with the uncontrolled EOMs. This means there will be CV coasting without thruster input $\rightarrow$ ($a_r = a_s = a_z = 0$) Which yields uncontrolled EOMs of the form:

$$\frac{d^2r}{dt^2} - 2n\frac{ds}{dt} - 3n^2r = 0$$
$$\frac{d^2s}{dt^2} + 2n\frac{dr}{dt} = 0$$
$$\frac{d^2z}{dt^2} + n^2z = 0$$

2. Initial values of the TV can be found Swift Observatory TLE history. SPEAR indents to rendezvous with Swift Observatory on August 22, 2026; so, values from that day will be used.
   - $R_{TV} = 6721.464$ km
3. Changing the along-track initial condition will simulate a V-bar approach; while changing the radial-track initial condition will simulate a R-bar approach.
   - $s_0 = -1000$ m: V-bar approach. For an uncontrolled simulation, the CV shares the same exact orbital period and velocity. This will result in the CV simply coasting and trailing the TV.
   - $r_0 = -1000$ m: R-bar approach. For an uncontrolled simulation, the CV has a faster orbital rate than the TV. This will result in the CV simply drifting past the TV.
4. What about placing the CV at an "above and ahead" location to the TV?
   - Higher orbit means the CV will be at a slower mean orbital rate than the TV. Because the CV is moving slower, it will naturally drift backward relative the the TV along-track axis. This allows the TV to naturally catch up to the CV from below.
   - The CV can enter a waiting period to allow the proper phasing to develop before executing the next maneuver.

Uncontrolled CWH Trajectory Simulation

Givens

- $\mu = 3.986004356 \times 10^{14}$ m^3/s^2
- $R_E = 6378.1366 \times 10^3$ m
- $R_{TV} = 6721.464 \times 10^3$ m

CV Initial Conditions

values subject to change

- $r_0 = 0$ or -1000 m
- $s_0 = 0$ or -1000 m
- $z_0 =$ m
- $\dot{r}_0 = 0$ or -1.0 m/s
- $\dot{s}_0 = 0$ or -1.0 m/s
- $\dot{z}_0 = 0$ m/s

Equations

$$n = \sqrt{\frac{\mu}{R_{TV}^3}} \quad \rightarrow \quad n = 0.0011 \text{ m/s}$$

$$T_{orbit} = \frac{2\pi}{n} \quad \rightarrow \quad T_{orbit} = 5484.11811 \text{ sec}$$

Results

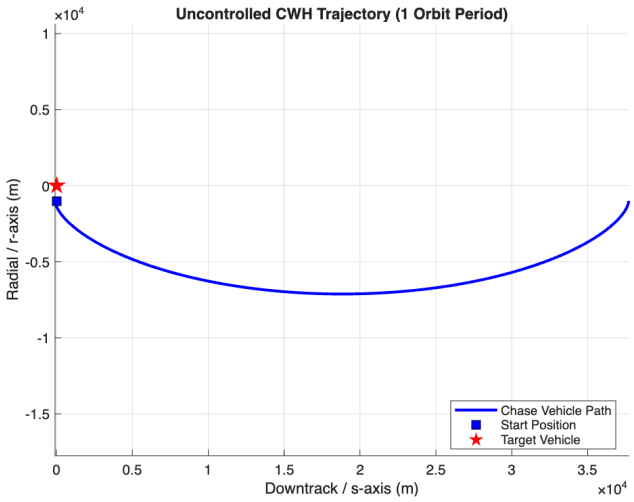


Figure 1: R-bar approach with initial conditions: $r_0 = -1000$ m and $\dot{r}_0 = 1.0$ m/s

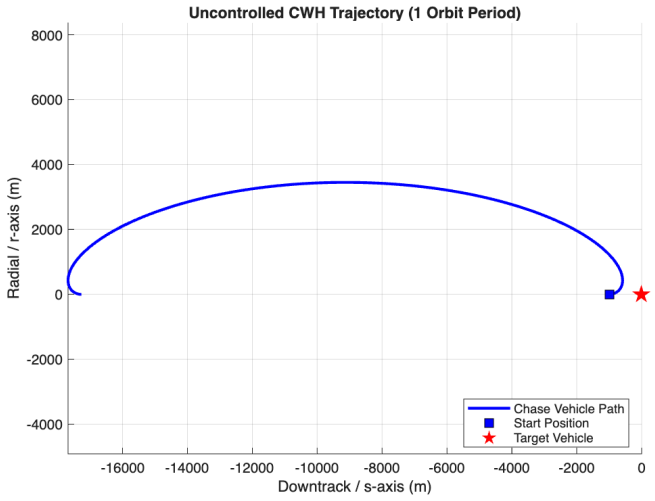


Figure 2: V-bar approach with initial conditions: $s_0 = -1000$ m and $\dot{s}_0 = 1.0$ m/s

Script Listing

Listing 1: CWH Trajectory without a Controller

```
% Elijah Bass
% AAE 450 - Orbit Boost
% Uncontrolled CWH Relative Motion Simulation

clear; clc; close all;

%% Given Orbital Parameters
mu = 3.986004356e14; % [m^3/s^2] Earth's gravitational parameter
R_TV = 6721.464e3; % [m] Target vehicle radius

%% Calculated Orbital Parameters
n = sqrt(mu / R_TV^3); % [rad/s] mean orbital rate
T_orbit = 2*pi/n; % [s] orbital period

%% Initial Conditions
r0 = -1000; % [m] radial track axis position
s0 = 0; % [m] along-track axis position
z0 = 0; % [m] cross-track axis position
rdot0 = 1.0; % [m/s] radial track axis velocity
sdot0 = 0; % [m/s] along-track axis position
zdot0 = 0; % [m/s] radial track axis position

%% Simulation Setup
X0 = [r0; s0; z0; rdot0; sdot0; zdot0]; % initial condition vector
t_span = [0, T_orbit]; % simulate for 1 orbital period
options = odeset('RelTol', 1e-8, 'AbsTol', 1e-8); % ode tolerances
[t, X_out] = ode45(@(t, X) cwh_unforced(t, X, n), t_span, X0, options);

%% Plotting
figure(1)
hold on; grid on;
plot(X_out(:,2), X_out(:,1), 'b-', 'LineWidth', 2, 'DisplayName', 'Chase Vehicle Path');
plot(s0, r0, 'ks', 'MarkerFaceColor', 'b', 'MarkerSize', 8, 'DisplayName', 'Start Position');
plot(0, 0, 'rp', 'MarkerFaceColor', 'r', 'MarkerSize', 12, 'DisplayName', 'Target Vehicle');
title('Uncontrolled CWH Trajectory (1 Orbit Period)');
xlabel('Downtrack / s-axis (m)');
ylabel('Radial / r-axis (m)');
legend('Location', 'best');
axis equal; hold off;
```

```
40
41 %% Unforced CWH Equations Function
42 function dX = cwh_unforced(~, X, n)
43
44     r = X(1); s = X(2); z = X(3);
45     rdot = X(4); sdot = X(5); zdot = X(6);
46
47     rddot = 2 * n * sdot + 3 * n^2 * r;
48     sddot = -2 * n * rdot;
49     zddot = -n^2 * z;
50
51     dX = [rdot; sdot; zdot; rddot; sddot; zddot];
52 end
```

Controller Development

Equations of Motion

$$\frac{d^2r}{dt^2} - 2n\frac{ds}{dt} - 3n^2r = a_r$$
$$\frac{d^2s}{dt^2} + 2n\frac{dr}{dt} = a_s$$
$$\frac{d^2z}{dt^2} + n^2z = a_z$$

a_r, a_s, a_z represent the control accelerations (thrust divided by mass) applied by the chase vehicle.

Error Dynamics

The goal is to get the CV to approach a desired state. This location will be defined as 30 m behind the TV on the along-track axis.

$$\boldsymbol{X}_{ref} = \begin{bmatrix} r_{ref} \\ s_{ref} \\ z_{ref} \end{bmatrix} \qquad \dot{\boldsymbol{X}}_{ref} = \begin{bmatrix} \dot{r}_{ref} \\ \dot{s}_{ref} \\ \dot{z}_{ref} \end{bmatrix} = 0$$

Positional error signals will be fed into a compensator defined by the following equations. The target state is a fixed point so the rate error are just the current relative velocity.

$$e_r = r - r_{ref} \qquad \dot{e}_r = \dot{r}$$
$$e_s = s - s_{ref} \qquad \dot{e}_s = \dot{s}$$
$$e_z = z - z_{ref} \qquad \dot{e}_z = \dot{z}$$

Control Law Formulation

Position and velocity control of the spacecraft is needed; therefore, a PD controller will be used. The compensator will calculate the required thrust acceleration as a combination of position and rate error.

$$a_r = -K_{p,r}e_r - K_{d,r}\dot{e}_r$$
$$a_s = -K_{p,s}e_s - K_{d,s}\dot{e}_s$$
$$a_z = -K_{p,z}e_z - K_{d,z}\dot{e}_z$$

Closed Loop Dynamics

Z-Axis:

$$\ddot{z} + n^2 z = -K_{p,z} e_z - K_{d,z} \dot{e}_z$$

The target $z_{ref} = 0$ leads $e_z = z$ and $\dot{e}_z = \dot{z}$ This leads the equation to become a damped simple-harmonic oscillator:

$$\ddot{z} + K_{d,z}\dot{z} + \left(n^2 + K_{p,z}\right) z = 0$$

R & S-Axes:

$$\ddot{r} + K_{d,r}\dot{r} + K_{p,r} - \underbrace{2n\dot{s}}_{\text{Coriolis}} - \underbrace{3n^2 r}_{\text{centrifugal}} = 0$$

$$\ddot{s} + K_{d,s}\dot{s} + K_{p,s} - \underbrace{2n\dot{s}}_{\text{Coriolis}} = 0$$

Gain Tuning

To determine the numerical value for the gains, the characteristic equation must be analyzed. For a second-order system, the characteristic equation is written:

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

Using the z-axis damped simple-harmonic oscillator:

$$2\zeta\omega_n = K_{d,z}$$

$$\omega_n^2 = n^2 + K_{p,z}$$

We can use settling time of 600 s and max overshoot of $< 5\%$ to calculate the natural frequency and and damping ratio.

$$M_p = \exp\left(-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}\right) < 5\% \quad \rightarrow \quad \zeta = 0.6901$$

$$t_s = \frac{4}{\zeta\omega_n} \quad \rightarrow \quad \omega_n = 0.00966 \text{ rad/s}$$

Substituting these values into the damped simple-harmonic oscillator result in the following gains:

$$K_{p,z}$$

GFCS GOLD Rules Constraint

GFCS GOLD Rules 1.30 constrain the speed at which the controller can act. This is because of the

References

[1] Michael D. Griffin and James R. French. *Space Vehicle Design*. 2nd. Reston, VA: American Institute of Aeronautics and Astronautics (AIAA), 2004. ISBN: 1-56347-539-1.

[2] ISS Tracker. *SWIFT - Satellite Information*. Accessed: Nov. 6, 2026. 2026. URL: https://isstracker.pl/en/satellites/28485 (visited on 11/06/2026).

[3] Mai Bando and Akira Ichikawa. "In-Plane Motion Control of Hill–Clohessy–Wiltshire Equations by Single Input". In: *Journal of Guidance, Control, and Dynamics* 36.5 (2013), pp. 1512–1522. DOI: 10.2514/1.57197. eprint: https://doi.org/10.2514/1.57197. URL: https://doi.org/10.2514/1.57197.

[4] NASA Goddard Space Flight Center. *Goddard Space Flight Center rules for the design, development, verification, and operation of flight systems*. Tech. rep. GSFC-STD-1000 Rev. G. National Aeronautics and Space Administration, 2021. URL: https://soma.larc.nasa.gov/lws/pdf_files/3.11%20GSFC-STD-1000Revg_signature.pdf.